

About Ops Center Administrator

Ops Center Administrator is a module of Ops Center that will allow you to easily manage Hitachi Storage systems in the enterprise including both SAN and NAS. One can easily provision storage to servers including the FC zoning on the SAN switches. One can also protect the volumes using either local replications (Snapshots) or remote replications. On the NAS side, one can provision shares and NFS exports.

Launching Administrator

From the Ops Center portal, one can click on the administrator link to launch the administrator.

The Administrator dashboard will be displayed.



1 Storage
Systems



3 Servers



0 Fabric
Switches



1 Virtual
Storage
Machines

Capacity Alerts



Data Protection Alerts



Jobs Alerts



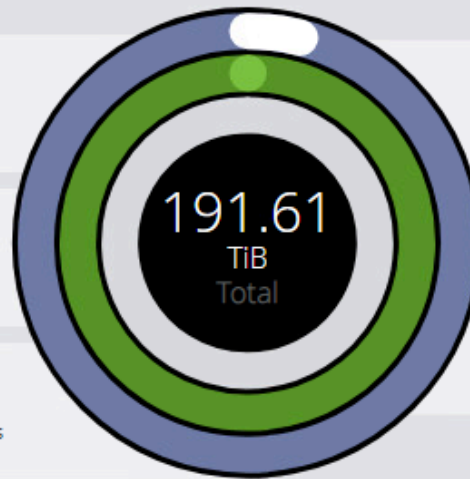
Hardware Alerts



- ☐ 4% Subscribed Capacity
- ☒ 191.61 TiB Physical Capacity

- ☒ 184.49 GiB Thin Used
- ☒ 191.43 TiB Thin Free

- ☐ 191.61 TiB Allocated to Pools
- ☐ 3.17 GiB Unallocated to Pools



Tier Breakdown

Tiers Protection Savings

- ☒ Diamond
- ☒ Platinum
- ☒ Gold
- ☒ Silver
- ☒ Bronze
- ☒ External

From the dashboard, one can easily see the overview status of the storage systems in the enterprise. A single instance of the Administrator module can support 50 Storage systems, 10,000 servers, and 1.5 million volumes.

Provisioning Storage

We will go through the steps to provision storage and create a snapshot schedule for the newly provisioned storage

On the left side, there is a panel with Storage Systems, Servers, Fabric Switches, Virtual File Servers, and Virtual Storage Machines.

Click on the servers.

The following is displayed.

HITACHI

Ops Center Administrator

Dashboard

Jobs

Monitoring

Dashboard

Servers

Servers

Server Groups

Actions...

	ID	ALERTS	SERVER NAM...	IP ADDRESS ...	PROTOCOL	WWN/ISCSI I...	OS TYPE	VOLUME COU...	REPLICATION...	STORAGE SYS...
	1	—	UCS-Profil...	—	Fibre	20:00:00:2...	VMWARE EX	3	—	745210
	2	—	UCS-Profil...	—	Fibre	20:00:00:2...	VMWARE EX	3	—	745210
	3	—	UCS-profil...	—	Fibre	20:00:00:2...	VMWARE EX	3	—	745210

3 items 0 selected

show

200

▼

items

«

1

»

From this window, one can add and modify servers and server groups.

The lab environment has 3 vmware servers.

Select a server and click on the Actions... button and one should see the following window.

HITACHI | Ops Center Administrator Dashboard Jobs Monitoring   

Dashboard > Servers ▾

Servers

► Summary

Servers

Server Groups

   Actions...



Attach Volumes ▸

Use Existing LUN Paths

Attach Existing Volumes

Create, Attach and Protect Volumes with Local Replication

Create, Attach and Protect Volumes with High Availability

ID								VOLUME COU...	REPLICATION...	STORAGE SYS...
1	—	UCS-Pr					EX	3	—	745210
2	—	UCS-Profil...	—	Fibre	20:00:00:2...	VMWARE EX	3	—	—	745210
3	—	UCS-profil...	—	Fibre	20:00:00:2...	VMWARE EX	3	—	—	745210

3 items 1 selected

show 200 ▾ items « 1 »

Select the second option, which is "Create, Attach and Protect Volumes with Local Replication".

The following window will appear.

HITACHI | Ops Center Administrator

DashboardJobsMonitoring

Dashboard > Servers > Create, Attach and Protect Volumes with Local Replication

Create, Attach and Protect Volumes

1. Create Volumes2. Attach Settings3. Create Paths4. Protect Volumes5. Operation Plan

STORAGE SYSTEM

VSP E Series (745210) ▾

SUBSCRIBED CAPACITY

4% (8.15 TiB / 191.61 TiB)

VIRTUAL STORAGE MA...

No VSM Selected (Defc ▾

For servers or server groups that have mis-matched Host Modes or Host Mode Options, please use the "Attach Like Volumes" function from the server's volume inventory, in order to prepopulate paths properly.

VOLUME LABEL

label

LABEL SUFFIX

e.g. 1

NUMBER OF VOLUMES

1

SIZE

1

GiB ▾

VOLUME ID/RANGE

Auto ▾

From

-

To

VIRTUAL ID/RANGE

Auto ▾

From

-

To

POOL TYPE

Thin ▾

POOL TIER

Platinum ▾

POOL

Auto Selected ▾

TIERING POLICY

-

CAPACITY SAVING

No ▾

Cancel

Next

On this window, select the storage system, enter the volume label, label suffix, number of volumes, and size.

For the storage system, just leave the default.

For the volume label, enter your initials plus vol (i.e. HKvolume)

Label Suffix, enter 1.

Number of volumes, enter 1

for the size, enter 1 and leave GiB selected.

After doing that, click on the + .

The following window will appear.



Create, Attach and Protect Volumes

1. Create Volumes

2. Attach Settings

3. Create Paths

4. Protect Volumes

5. Operation Plan

STORAGE SYSTEM

VSP E Series (745210) ▾

SUBSCRIBED CAPACITY

4% (8.15 TiB / 191.61 TiB)

VIRTUAL STORAGE MA...

No VSM Selected (Def: ▾)

For servers or server groups that have mismatched Host Modes or Host Mode Options, please use the "Attach Like Volumes" function from the server's volume inventory, in order to prepopulate paths properly.

VOLUME LABEL

hkvolume

LABEL SUFFIX

e.g. 1

NUMBER OF VOLUMES

1

SIZE

1

GiB ▾



VOLUME ID/RANGE

Auto ▾

From

-

To

VIRTUAL ID/RANGE

Auto ▾

From

-

To

POOL TYPE

Thin ▾

POOL TIER

Platinum ▾

POOL

Auto Selected ▾

TIERING POLICY

-

CAPACITY SAVING

No ▾

VOLUME LABEL

hkvolume

LABEL SUFFIX

NUMBER OF VOLUMES

1

SIZE

1

GiB ▾



VOLUME ID/RANGE

Auto ▾

-

VIRTUAL ID/RANGE

Auto ▾

-

POOL TYPE

Thin ▾

POOL TIER

Platinum ▾

POOL

Auto Selected ▾

TIERING POLICY

-

CAPACITY SAVING

No ▾

Cancel

Next

click on the next button at the bottom.

The following window will display.

HITACHI | Ops Center Administrator

DashboardJobsMonitoring

Dashboard > Servers > Create, Attach and Protect Volumes with Local Replication

Create, Attach and Protect Volumes

1. Create Volumes2. Attach Settings3. Create Paths4. Protect Volumes5. Operation Plan

STORAGE SYSTEM
VSP E Series (745210)

HOST MODE
VMWARE EX

HOST MODE OPTIONS
2 - VERITAS Database ...

HOST GROUP NAME
(Optional)

MANDATE LUN ALIGN...
YesNo

AUTO CREATE ZONE
YesNo

View Proposed Volume ID Selection

NUMBER O...	LABEL	LABEL SUF...	SIZE	POOL TYPE	TIER	POOL
1	hkvolume	1	1.00 GiB	Thin	Platinum	Auto Sel...

CancelPreviousNext

One can enter the host mode, host mode option, host group name (optional), LUN Alignment option, auto create zone option.

If the Auto Create Zone option is selected, the zones will be created in the SAN fabric.

Just keep the default selections and click next.

The following window will be displayed.



Create, Attach and Protect Volumes

1. Create Volumes

2. Attach Settings

3. Create Paths

4. Protect Volumes

5. Operation Plan

Switch View

Suggest

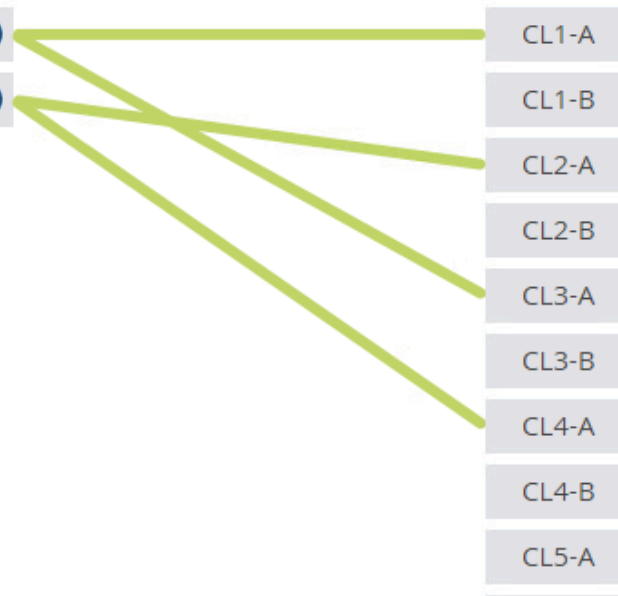
 Existing New

1 Servers

16 Target Storage Ports

UCS-profile-1 (VMWARE EX)

20:00:00:25:B5:54:0A:00	
20:00:00:25:B5:54:0B:00	



Cancel

Previous

Next

This window allows one to make the connections between the server and the storage system. The existing connections show up in orange. One can add additional connections, and the new connections will show up in blue.

Do not add new connections,click next and the following window is displayed



Create, Attach and Protect Volumes

1. Create Volumes

2. Attach Settings

3. Create Paths

4. Protect Volumes

5. Operation Plan

REPLICATION TYPE

Snap

None

SNAP RETENTION POLICY (MAX 1024)

Snap Retention

CONSISTENCY

Yes

No

Note: For volumes that already contain one or more snapshots, new snapshots will be saved in the same pool as the existing snapshots.

Hourly

Daily

Weekly

Monthly

START MINUTE

0

EVERY

1

Hours

▲ Snap will start every 1 hours of the UTC time. Current time in UTC is 07/17/23 02:17 PM.

USE REPLICATION GROUP

Use New



REPLICATION GROUP NAME

Replication Group Name

POOL FOR SNAPSHOT

STORAGE SYSTEM ID

STORAGE POOL

Cancel

Previous

Next

Submit



From this window, one can configure the data protection.

If one selects, None on Replication Type, the volume(s) will not be protected.

If one selects, Snap on Replication Type, one can specify the schedule for the snapshot. (keep the default entry)

Below the Replication type selection, there is "Snap Retention Policy".

Please enter 4. This setting sets the number of snapshots to retain.

One can also select consistency option. If there are multiple volumes, the group of volumes will create a snapshot at the same time.

Below the Schedule, "Replication Group Name". One can use an existing one by selecting one or create a new one by entering a name. Create a new one and enter your initials-RGN. (i.e. HK-RGN)

Keep the default for the rest of the entries.

Click submit to create, attach, and protect volumes.